Computational Finance Assignment 2

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Throughout this assignment we price the European basket call

$$V(0) = e^{-rT}\,\mathbb{E}^{\mathbb{Q}}\big[\max\big(w_1S_1(T) + w_2S_2(T) - K,\, 0\big)\big]\,, \tag{1}$$

with $n = 2$ underlyings, weights $w_1 = w_2 = \frac{1}{2}$, maturity $T = 1$, risk-free rate $r = 0.02$, no dividends and strike $K = 100$ (at-the-money).

The parameters (in particular the drift $\mu = 0.10$) are given in the physical measure $\mathbb{P}$, but a no-arbitrage price must be computed under the risk-neutral measure $\mathbb{Q}$. We make the change of measure from $\mathbb{P}$ to $\mathbb{Q}$ explicit, since it is what determines the dynamics we simulate.

For the risk-neutral measure, we take the money-market account $B(t) = e^{rt}$ (with $B(0) = 1$) as numéraire. By the first fundamental theorem of asset pricing, the absence of arbitrage is equivalent to the existence of a measure $\mathbb{Q}$ equivalent to $\mathbb{P}$ under which the discounted price of every tradable asset is a martingale:

$$\frac{S_i(t)}{B(t)} = e^{-rt}S_i(t) \quad \text{is a } \mathbb{Q}\text{–martingale}, \qquad i = 1, 2.$$

This condition fixes the drift of each tradable asset under $\mathbb{Q}$ to be rS_i, regardless of its physical drift, and the change of measure is realized through Girsanov's theorem, which changes the drifts but leaves volatilities and the correlation structure unchanged.

For the log-normal underlying assets, note that under $\mathbb{P}$ each is a geometric Brownian motion $dS_i = \mu\, S_i\, dt + \sigma_i S_i\, dW_i^{\mathbb{P}}$. Defining the market price of risk $\hat{r}_i = (\mu - r)/\sigma_i$ and the shifted process $W_i^{\mathbb{Q}}(t) = W_i^{\mathbb{P}}(t) + \hat{r}_i\, t$, by Girsanov's theorem we have that $W_i^{\mathbb{Q}}$ is a $\mathbb{Q}$-Brownian motion. Note that this procedure was already done in another set of exercises/assignment. Substituting $dW_i^{\mathbb{P}} = dW_i^{\mathbb{Q}} - \hat{r}_i\, dt$,

$$\begin{aligned} dS_i &= \mu\, S_i\, dt + \sigma_i S_i\big(dW_i^{\mathbb{Q}} - \hat{r}_i\, dt\big) = \big(\mu - \sigma_i\hat{r}_i\big)S_i\, dt + \sigma_i S_i\, dW_i^{\mathbb{Q}} \\ &= r\, S_i\, dt + \sigma_i S_i\, dW_i^{\mathbb{Q}}, \end{aligned}$$

so the drift becomes r while the volatility σ_i is untouched. Notice that $d\big(e^{-rt}S_i\big) = e^{-rt}\sigma_i S_i\, dW_i^{\mathbb{Q}}$ has no drift term, confirming the martingale property. Because Girsanov does not change the covariation $d\langle W_1, W_2\rangle = \rho\, dt$, the correlation ρ is the same under $\mathbb{P}$ and $\mathbb{Q}$, so we may use the given ρ directly.

In Question 1(b) the second asset follows a square-root process under $\mathbb{P}$, $dS_2 = \kappa(\theta - S_2)\,dt + \sigma_2\sqrt{S_2}\,dW_2^{\mathbb{P}}$. The important point is that S_2 is itself a tradable basket underlying, so the same no-arbitrage condition forces $e^{-rt}S_2$ to be a $\mathbb{Q}$-martingale and hence its $\mathbb{Q}$-drift to be rS_2. Writing $W_2^{\mathbb{Q}}(t) = W_2^{\mathbb{P}}(t) + \int_0^t \gamma(s)\,ds$ with the market price of risk

$$\gamma(t) = \frac{\kappa(\theta - S_2(t)) - r\,S_2(t)}{\sigma_2\sqrt{S_2(t)}},$$

Girsanov's theorem absorbs the entire physical drift and yields

$$dS_2(t) = r\,S_2(t)\,dt + \sigma_2\sqrt{S_2(t)}\,dW_2^{\mathbb{Q}}(t),$$

which is exactly the square-root process of the technical hint. Note that the mean-reversion parameters κ, θ drop out entirely. We use these risk-neutral dynamics in every implementation below.

1 Basket Options via Monte Carlo Simulation

(a) Both underlyings log-normal

Note that under $\mathbb{Q}$ both assets are geometric Brownian motions with drift r, then we can write

$$S_i(T) = S_i(0)\exp\Big(\big(r - \tfrac{1}{2}\sigma_i^2\big)T + \sigma_i\sqrt{T}\,Z_i\Big), \qquad i = 1, 2,$$

where (Z_1, Z_2) are standard normals with correlation $\rho = 0.30$. Since the solution at maturity is known in closed form, we do not need time discretizations and the simulation is *exact*. We draw two independent standard normals $Z_1, \widetilde{Z}$ and correlate them with a Cholesky decomposition, $Z_2 = \rho Z_1 + \sqrt{1-\rho^2}\,\widetilde{Z}$. This comes from the fact that $dW_1 dW_2 = \rho dt$. We can rewrite this condition using a covariance matrix Σ:

$$\Sigma = \begin{pmatrix} dt & \rho \cdot dt \\ \rho \cdot dt & dt \end{pmatrix} = dt\begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}.$$

Then, we just apply Cholesky decomposition with $\Sigma = LL^T$, where we find that

$$L = \begin{pmatrix} 1 & 0 \\ \rho & \sqrt{1-\rho^2} \end{pmatrix}.$$

Finally, to find the correlated variables, we just multiply the Cholesky matrix L by a vector of independent standard normal variables scaled by time, $\begin{pmatrix} \sqrt{dt}Z_1 \\ \sqrt{dt}\widetilde{Z} \end{pmatrix}$. After the computations, we get $dW_1 = \sqrt{dt}Z_1$ and $dW_2 = \rho dW_1 + \sqrt{1-\rho^2}\sqrt{dt}\widetilde{Z}$.

For M paths, with discounted payoffs $Y^{(m)} = e^{-rT} \max\big(w_1 S_1^{(m)} + w_2 S_2^{(m)} - K, 0\big)$, the Monte Carlo estimator and its standard error are

$$\widehat{V}(M) = \frac{1}{M} \sum_{m=1}^{M} Y^{(m)}, \qquad \widehat{\mathrm{SE}}(M) = \frac{s(M)}{\sqrt{M}},$$

with $s(M)$ the sample standard deviation of the discounted payoffs, and the 95% confidence interval is $\widehat{V}(M) \pm 1.96\,\widehat{\mathrm{SE}}(M)$.

At $M = 10^6$ paths, for example, we obtain

$$\widehat{V} = 9.04661, \qquad \widehat{\mathrm{SE}} = 0.01426, \qquad 95\%\ \mathrm{CI} = [\,9.01866,\ 9.07455\,].$$

For the convergence test, we know that a correctly implemented Monte Carlo estimator has an error that decays as $\mathcal{O}(M^{-1/2})$ by the central limit theorem, so the standard error is a straight line of slope $-\frac{1}{2}$ in log-log scale. Figure 1 confirms this and the fitted slope is -0.498, in agreement with the theoretical rate. Figure 2 shows the standard error along the $1/\sqrt{M}$ reference line.

M	price	SE	95% CI half-width
2000	9.37199	0.32213	0.63138
5000	8.88720	0.19369	0.37964
10000	9.06643	0.14257	0.27944
25000	8.92674	0.08906	0.17456
50000	8.98641	0.06331	0.12409
100000	9.02143	0.04474	0.08768
250000	9.06261	0.02861	0.05607
500000	9.06850	0.02019	0.03956
1000000	9.04661	0.01426	0.02794

convergence rate (slope of log SE vs log M) = -0.498 (expected -0.5)

Figure 1: Monte Carlo price, standard error and CI half-width. Fitted convergence rate (slope of $\log \widehat{\mathrm{SE}}$ vs $\log M$).

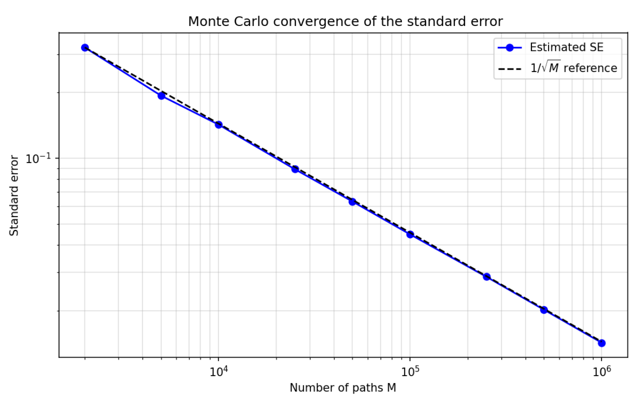

Figure 2: Standard error versus M in log-log scale, with the $1/\sqrt{M}$ reference line.

(b) One log-normal, one CIR

Observe that the first asset is log-normal, so we step it with its exact GBM solution and therefore, we do not have any discretization error. The second asset follows the square-root process $dS_2 = rS_2\,dt + \sigma_2\sqrt{S_2}\,dW_2$ with $\sigma_2 = 1.5$. In here the exact non-central chi-square law of the square-root process (the hint given at the beginning of the assessment) only applies to the marginal of S_2, not jointly with a correlated S_1, and since the two Brownian motions are correlated ($\rho = 0.1$) exact joint simulation is not available. We therefore discretize S_2 with the Euler-Maruyama scheme explained in Lecture 4. For a general SDE $dX_t = \mu(t, X_t)\,dt + \sigma(t, X_t)\,dW_t$ on the grid $t_k = kh$, $k = 0, \ldots, n$, $nh = T$, the scheme reads

$$\widehat{X}_{t_k} = \widehat{X}_{t_{k-1}} + \mu\big(t_{k-1}, \widehat{X}_{t_{k-1}}\big)\,h + \sigma\big(t_{k-1}, \widehat{X}_{t_{k-1}}\big)\sqrt{h}\,Z_k, \qquad Z_k \sim N(0,1) \text{ i.i.d.}$$

For S_2 we have $\mu(t, X) = rX$ and $\sigma(t, X) = \sigma_2\sqrt{X}$. The only modification we make is a full truncation $X \mapsto X^+ := \max(X, 0)$ inside the square root (and drift), which keeps the diffusion $\sigma_2\sqrt{X}$ real-valued, since the unmodified scheme can drive a path below zero. Under $\mathbb{Q}$ the square-root process has no mean reversion (κ, θ dropped out), so the condition $2 \cdot 0 \geq \sigma_2^2$ fails and the origin is attainable in principle. Then, the full truncation is therefore needed to keep the scheme well defined. The step is therefore

$$\widehat{S}_2(t_k) = \widehat{S}_2(t_{k-1}) + r\,\widehat{S}_2^+(t_{k-1})\,h + \sigma_2\sqrt{\widehat{S}_2^+(t_{k-1})}\,\sqrt{h}\,Z_k^{(2)},$$

with $\widehat{S}_2^+ := \max(\widehat{S}_2, 0)$, and at maturity we set $S_2(T) = \max(\widehat{S}_2, 0)$. The normal $Z_k^{(2)} = \rho\,Z_k^{(1)} + \sqrt{1-\rho^2}\,\widetilde{Z}_k$ is correlated through Cholesky decomposition. At $n = 200$ steps and $M = 400\,000$ paths we obtain

$$\widehat{V} = 7.02958, \qquad \widehat{\mathrm{SE}} = 0.01663, \qquad 95\%\ \mathrm{CI} = [\,6.99697, 7.06218\,].$$

We do two different convergence tests. The first one is the time discretization error. Lecture 4 slides distinguish two measures of the discretisation error. A scheme has strong order β if $\mathbb{E}\big[\,|\widehat{X}_{nh} - X_T|\,\big] \leq c\,h^\beta$, which controls the pathwise distance, and weak order β if $\big|\mathbb{E}[f(\widehat{X}_{nh})] - \mathbb{E}[f(X_T)]\big| \leq c\,h^\beta$ for all sufficiently small h and all smooth $f \in C_P^{2\beta+2}$ (functions whose derivatives up to order $2\beta+2$ have polynomial growth). The Euler-Maruyama scheme has strong order $\beta = \frac{1}{2}$ and weak order $\beta = 1$. For option pricing only the expected discounted payoff matters, so the relevant quantity is the weak error with f the basket-call payoff $f(S) = e^{-rT}\max(w_1S_1 + w_2S_2 - K, 0)$.

We estimate this weak error while eliminating the Monte Carlo noise by reusing the same Brownian increments across all grids. We simulate S_2 on a fine grid of 256 steps and build each coarser grid by adding the fine increments into blocks, so the only difference between grids is the discretization of S_2. Measured against the fine reference 7.04501, each halving of h roughly halves the bias (Figure 3), and the fitted slope of $\log|\text{bias}|$ versus $\log h$ over the resolved

```
 reference price (256 steps, M = 720000): 7.04512
n_steps      price     |bias|
-----------------------------
      4    7.03762    0.00751
      8    7.04156    0.00356
     16    7.04336    0.00177
     32    7.04425    0.00087
     64    7.04471    0.00041
    128    7.04501    0.00012

 weak convergence order (slope of log|bias| vs log h) = 1.03  (expected 1.0)
```

Figure 3: Time-discretisation bias against the 256-step reference 7.04501 under common random numbers. Fitted weak order: 1.03.

grids is 1.03, which is an empirical weak order close to $\beta = 1$. Figure 4 shows the bias against the order-1 reference line.

The second convergence test is the simulation error. Fixing the grid at $n = 200$ steps and varying M, the standard error again decays as $1/\sqrt{M}$ (Figure 5). Note that the fitted slope is -0.497, matching the $\mathcal{O}(M^{-1/2})$ Monte Carlo rate. Figure 6 shows the decay.

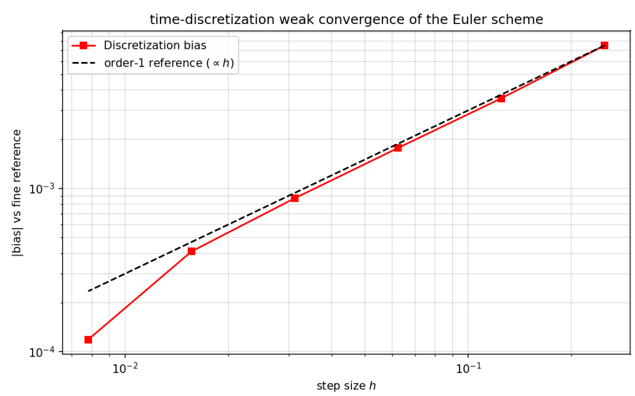

Figure 4: Time-discretisation weak convergence of the Euler-Maruyama scheme, with the order-1 reference $\propto h$.

```
       M      price         SE   95% CI half-width
--------------------------------------------------
    5000    6.83851    0.14507             0.28434
   10000    6.91814    0.10473             0.20527
   25000    7.02659    0.06731             0.13193
   50000    7.01786    0.04721             0.09252
  100000    6.99517    0.03300             0.06467
  200000    7.00861    0.02346             0.04598
  400000    7.02958    0.01663             0.03260

 convergence rate (slope of log SE vs log M) = -0.497  (expected -0.5)
```

Figure 5: Simulation error at fixed grid $n = 200$. Fitted convergence rate (slope of $\log\widehat{\mathrm{SE}}$ vs $\log M$). The 95% CI = [6.99697, 7.06218]

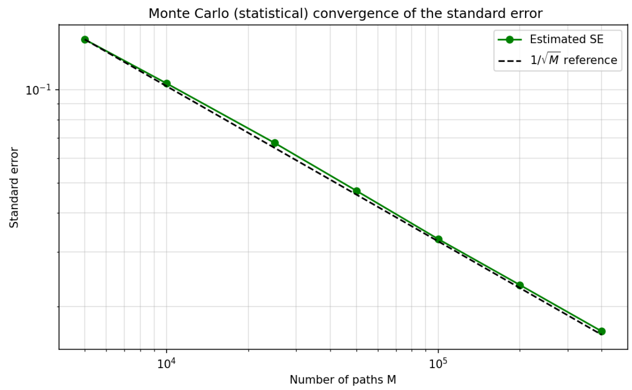

Figure 6: Monte Carlo convergence of the standard error, with the $1/\sqrt{M}$ reference line.

2 Basket Options via Moment Matching

The moment-matching method replaces the unknown distribution of the basket $B_T = w_1 S_1(T) + w_2 S_2(T)$ by an analytically tracFigure distribution whose first few moments coincide with those of B_T, and prices the call in closed form (or by one-dimensional quadrature) under the approximating distribution. We match two, three and four moments and examine how the approximation error behaves as more moments are matched.

Moments of the basket

We need the raw moments $m_k = \mathbb{E}^{\mathbb{Q}}[B_T^k]$, which we expand with the binomial theorem,

$$m_k = \sum_{j=0}^{k} \binom{k}{j} w_1^j w_2^{k-j}\, \mathbb{E}\big[S_1(T)^j\, S_2(T)^{k-j}\big].$$

Log-normal case (a). For correlated log-normal underlyings the log-returns are jointly normal, so for any exponents j, m,

$$\mathbb{E}\big[S_1(T)^j S_2(T)^m\big] = \exp\Big(j\mu_1 + m\mu_2 + \tfrac{1}{2}\big(j^2 v_1 + m^2 v_2 + 2jm\, c_{12}\big)\Big),$$

with $\mu_i = \ln S_i(0) + (r - \frac{1}{2}\sigma_i^2)T$, $v_i = \sigma_i^2 T$ and $c_{12} = \rho\,\sigma_1\sigma_2 T$, using $\mathbb{E}[e^N] = e^{\,\mathrm{mean} + \frac{1}{2}\mathrm{var}}$.

CIR case (b). Here $\rho = 0$, so S_1 and S_2 are independent and the joint moments factorise into the product of marginal moments. For S_1 we reuse the log-normal formula above. For S_2 the technical hint gives $S_2(T) = c\,Z$ with $c = e^{rT}\sigma_2^2\tau(T)/4$, $\tau(T) = (1 - e^{-rT})/r$ and $Z \sim \chi_0'^2(\lambda)$, $\lambda = 4S_2(0)/(\sigma_2^2\tau(T))$. The non-central chi-square with zero degrees of freedom has cumulants $\kappa_n =$

$2^{n-1}n!\,\lambda$, which we convert to raw moments through the standard cumulant-to-moment recursion $m_n = \sum_{k=0}^{n-1}\binom{n-1}{k}\kappa_{n-k}m_k$, and finally scale by c^k. From the moments we form the mean, variance, skewness γ_1 and kurtosis β_2 that the approximating distributions match. Their values are reported in Figures 7 and 8.

Approximating distributions

Now we determine the approximating distribution. In each case we use a member of the Johnson translation system, adding one parameter for every extra moment we match. We denote the matched central moments by the variance c_2, the skewness γ_1 and the kurtosis β_2.

As the lecture does, for the first two moments we fit a log-normal $B_T \sim \mathrm{LN}(\mu_L, \sigma_L^2)$ matching the expected basket value m_1 and the second moment m_2. The conditions $\mathbb{E}[B_T] = e^{\mu_L + \frac{1}{2}\sigma_L^2} = m_1$ and $\mathbb{E}[B_T^2] = e^{2\mu_L + 2\sigma_L^2} = m_2$ give in closed form

$$\sigma_L^2 = \ln\frac{m_2}{m_1^2}, \qquad \mu_L = \ln m_1 - \tfrac{1}{2}\sigma_L^2.$$

The basket call is then a call on a log–normal with forward $F = m_1$ and total variance σ_L^2, priced by the Black–style formula

$$C = e^{-rT}\big(m_1\Phi(d_1) - K\Phi(d_2)\big), \qquad d_1 = \frac{\ln(m_1/K) + \frac{1}{2}\sigma_L^2}{\sigma_L}, \quad d_2 = d_1 - \sigma_L.$$

For the three moments, a plain log-normal distribution cannot match a third moment, because once m_1 and m_2 are fixed its skewness and kurtosis are determined. We therefore add a location parameter and set $B_T = \eta + L$ with $L \sim \mathrm{LN}$, matching mean, variance and skewness. A shift does not change the shape, so B_T has the log-normal skewness $\gamma_1 = (\omega+2)\sqrt{\omega-1}$ with $\omega = e^{\sigma_L^2}$. Then, we invert this relation for ω numerically, which requires $\gamma_1 > 0$ (which is satisfied by both baskets). The variance fixes the scale through $\mathbb{E}[L] = \sqrt{c_2/(\omega-1)}$ and the mean fixes the shift $\eta = m_1 - \mathbb{E}[L]$. We determine that the call is then a log-normal call on L with the shifted strike $K - \eta$.

Finally, to control the kurtosis independently we move to the unbounded Johnson type, $B_T = \xi + \lambda\sinh\big((Z-\gamma)/\delta\big)$ with $Z \sim N(0,1)$ and four parameters $(\gamma, \delta, \xi, \lambda)$. Writing $\omega = e^{1/\delta^2}$ and $\Omega = \gamma/\delta$, its mean and variance are

$$\mathbb{E}[B_T] = \xi - \lambda\sqrt{\omega}\,\sinh\Omega, \qquad \mathrm{Var}(B_T) = \tfrac{1}{2}\,\lambda^2(\omega-1)\big(\omega\cosh 2\Omega + 1\big).$$

We use the fact that the standardized skewness and kurtosis depend only on the shape (δ, γ), not on the location/scale (ξ, λ). We therefore solve the 2×2 system for (δ, γ) from the target (γ_1, β_2). We evaluate this $2{\times}2$ system through Gauss–Hermite moments of *sinh* and solve it numerically. The call is priced by Gauss-Hermite quadrature.

(a) Both underlyings log-normal, $\rho = 0.30$

The basket moments are mean 102.02014, standard deviation 21.11290, skewness 0.69570 and kurtosis 3.92050. The Monte Carlo benchmark (exact simulation with antithetic variates, with the standard error taken from the 4×10^6 pair-averaged payoffs as the antithetic method requires) is 9.06113 with SE = 0.00390. This agrees with the Q1(a) estimate, which is a useful verification since both price the same $\rho = 0.30$ basket. Figure 7 shows that moving from two to three moments reduces the error by roughly a factor of three (from 0.04194 to 0.01319). The four-moment Johnson SU approximation does not provide any improvement in this case. A basket consisting of two correlated log-normal variables is itself very close to being log-normal, and the observed kurtosis of 3.92 is already well captured by the shifted log-normal approximation. As a result, the three-moment fit is accurate to within the level of Monte Carlo sampling error, while the additional kurtosis constraint imposed by the Johnson SU distribution leads to a slight overfitting of the tail behavior. Figure 9 (left) illustrates how the approximation error varies with the number of matched moments.

```
 basket moments: mean = 102.0201, std = 21.1129, skewness = 0.6957, kurtosis = 3.9205
 MC benchmark: 9.06113  (SE 0.00390)
moments matched      distribution      price   abs error
-----------------------------------------------------------
              2        log-normal    9.10307     0.04194
              3  shifted log-normal    9.07433     0.01319
              4        Johnson SU    9.08884     0.02770
```

Figure 7: Moment-matching prices and absolute error against the Monte Carlo benchmark 9.06113 (SE = 0.00390).

(b) One log-normal, one CIR, $\rho = 0$

The basket moments are mean 102.02014, standard deviation 15.02601, skewness 0.52780 and kurtosis 3.60990. The benchmark uses *exact* simulation: S_1 from its GBM solution and $S_2 = cZ$ with $Z \sim \chi_0'^2(\lambda)$ sampled exactly from the hint, giving 6.78609 with SE = 0.00304. Here the improvement is monotone (Figure 8). Notice how the error falls from 0.03895 (two moments) to 0.01433 (three) to 0.00460 (four), so each additional moment improves the approximation. Figure 9 (right) shows the monotone decay. Note that this price (≈ 6.78) is lower than the Q1(b) Monte Carlo price (≈ 7.02), which is expected because Question 2(b) sets $\rho = 0$ whereas Question 1(b) uses $\rho = 0.1$.

```
 basket moments: mean = 102.0201, std = 15.0260, skewness = 0.5278, kurtosis = 3.6099
 MC benchmark: 6.78609  (SE 0.00304)
moments matched      distribution      price   abs error
-----------------------------------------------------------
              2        log-normal    6.82504     0.03895
              3  shifted log-normal    6.80042     0.01433
              4        Johnson SU    6.79069     0.00460
```

Figure 8: Moment-matching prices and absolute error against the Monte Carlo benchmark 6.78215 (SE = 0.00303).

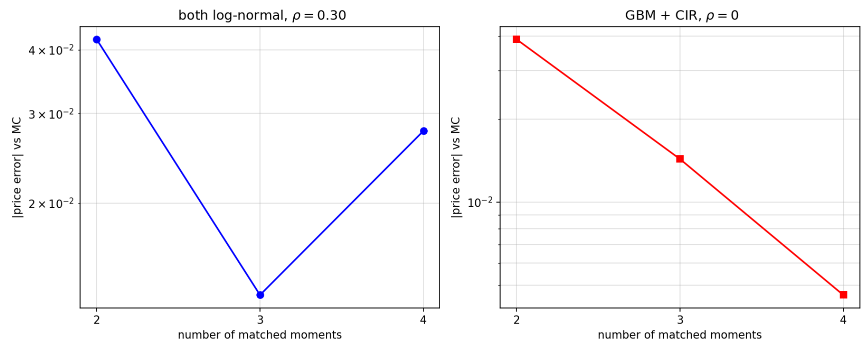

Figure 9: Absolute pricing error against the number of matched moments, for case (a) (left) and case (b) (right).

3 Basket Options via the COS Method

The basket call price is the discounted expectation of the payoff against the *joint* law of the two terminal prices,

$$V(0) = e^{-rT}\,\mathbb{E}^{\mathbb{Q}}\big[v(y_1,y_2)\big] = e^{-rT}\int_{a_1}^{b_1}\int_{a_2}^{b_2} v(y_1,y_2)\,f(y_1,y_2)\,dy_1\,dy_2, \qquad (2)$$

where (y_1, y_2) are two state variables chosen per case, f is their joint risk–neutral density, and $v(y_1,y_2) = \max\big(w_1 g_1(y_1) + w_2 g_2(y_2) - K, 0\big)$ is the payoff written through the maps g_i that send each state variable to its asset price. We use a two-dimensional expansion because the payoff couples $S_1(T)$ and $S_2(T)$ and the characteristic function of the basket sum $B_T = w_1S_1(T) + w_2S_2(T)$ has no closed form (a sum of correlated log-normals is intracFigure), whereas the marginal characteristic functions of the two terminal prices are known explicitly. This is why we use the 2D COS method of Ruijter and Oosterlee (2012).

The method has different steps. The first one is the 1D density recovery from the characteristic function. On a truncated interval $[a, b]$ a density g is represented by its Fourier-cosine series

$$g(y) \approx \sum_{k=0}^{N-1}{}' A_k \cos\Big(k\pi\tfrac{y-a}{b-a}\Big), \qquad A_k = \frac{2}{b-a}\int_a^b g(y)\cos\Big(k\pi\tfrac{y-a}{b-a}\Big)dy,$$

where the prime halves the $k = 0$ term. Defining $\omega_k = k\pi/(b-a)$ and extending the integral to $\mathbb{R}$ (the tails are negligible on a wide enough $[a, b]$), the coefficient follows from the characteristic function $\varphi(\omega) = \mathbb{E}[e^{iwY}] = \int g(y)e^{i\omega y}dy$:

$$\int_a^b g(y)\cos\big(\omega_k(y-a)\big)\,dy \approx \mathrm{Re}\big\{e^{-i\omega_k a}\,\varphi(\omega_k)\big\} =: \chi_k,$$

so that $A_k \approx \frac{2}{b-a}\,\chi_k$, using $\int_{\mathbb{R}} g(y)e^{i\omega y}dy = \varphi(\omega)$. We write the cosine basis as $u_k^{(i)}(y) := \cos\big(k\pi(y-a_i)/(b_i-a_i)\big)$ for the two dimensions $i = 1, 2$ introduced next.

The next step is the 2D expansion and the $\rho = 0$ factorization.The joint density is expanded on $[a_1, b_1] \times [a_2, b_2]$,

$$f(y_1, y_2) \approx \sum_{k_1}{}' \sum_{k_2}{}' A_{k_1,k_2}\, u_{k_1}^{(1)}(y_1)\, u_{k_2}^{(2)}(y_2),$$

with $A_{k_1,k_2} = \frac{2}{b_1-a_1}\frac{2}{b_2-a_2}\iint f\, u_{k_1}^{(1)} u_{k_2}^{(2)}\, dy$. Because $\rho = 0$ the state variables are independent, and therefore $f(y_1, y_2) = f_1(y_1)f_2(y_2)$, the joint characteristic function factorizes as $\varphi(\omega_1, \omega_2) = \varphi_1(\omega_1)\varphi_2(\omega_2)$, and consequently

$$A_{k_1,k_2} = \Big(\tfrac{2}{b_1-a_1}\chi_{k_1}\Big)\Big(\tfrac{2}{b_2-a_2}\chi_{k_2}\Big), \qquad \chi_{k_i} = \mathrm{Re}\{\varphi_i(\omega_{k,i})\, e^{-i\omega_{k,i}a_i}\}.$$

So, basically we only need the two marginal characteristic functions.

Next, we define the payoff cosine coefficients

$$V_{k_1,k_2} = \frac{2}{b_1-a_1}\frac{2}{b_2-a_2}\iint v(y_1, y_2)\, u_{k_1}^{(1)}(y_1)\, u_{k_2}^{(2)}(y_2)\, dy_1\, dy_2.$$

Substituting the expansion of f into (2), each integral $\iint v\, u_{k_1}^{(1)} u_{k_2}^{(2)}\, dy$ equals $\frac{1}{4}(b_1-a_1)(b_2-a_2)\, V_{k_1,k_2}$, and these factors cancel the $\frac{2}{b_i-a_i}$ inside A_{k_1,k_2}. Then, we can rewrite the price as

$$V(0) = e^{-rT}\sum_{k_1=0}^{N-1}{}' \sum_{k_2=0}^{N-1}{}' \chi_{k_1}\, \chi_{k_2}\, V_{k_1,k_2}\ .$$

Notice that the recovered density carries a factor $\frac{2}{b-a}$ per dimension, which is absorbed by the inverse transform, so χ_k goes without that factor while the payoff cosine coefficients V_{k_1,k_2} retains it.

Now, we need to compute this payoff coefficients. Notice that the basket payoff is not differentiable along the exercise boundary $w_1g_1(y_1)+w_2g_2(y_2) = K$, so the inner integrals have no clean closed form. We evaluate V_{k_1,k_2} with a composite trapezoidal rule on a grid $\{y_{1,m}\} \times \{y_{2,n}\}$. With $P_{mn} = v(y_{1,m}, y_{2,n})$ and cosine matrices $(C_i)_{m,k} = h_i\, \varpi_m\, u_k^{(i)}(y_{i,m})$ holding the trapezoidal weights ϖ_m and mesh h_i, all coefficients are obtained at once as

$$V = \frac{2}{b_1-a_1}\frac{2}{b_2-a_2}\, C_1^{\top} P\, C_2,$$

which is fully vectorized, so easier to implement in Python.

Moreover, as in the 1D case, we need to define the truncation domain. We fix each interval by the cumulant rule of the COS papers,

$$[a, b] = \Big[\kappa_1 - L\sqrt{\kappa_2 + \sqrt{\kappa_4}},\ \kappa_1 + L\sqrt{\kappa_2 + \sqrt{\kappa_4}}\Big], \qquad L = 12,$$

which centers the interval at the mean κ_1 and scales it by the standard deviation $\sqrt{\kappa_2}$ with a fourth-cumulant tail correction, so the domain adapts to each marginal automatically instead of being picked by hand. $L = 12$ makes the domain-truncation error negligible against the series error.

Finally, notice that for a smooth density whose characteristic function decays rapidly, the COS series-truncation error decays exponentially in N (Fang and Oosterlee, 2008), which is the expected rate. The Monte Carlo benchmark carries a statistical error of 10^{-3} that would mask it, so we measure the error against a converged COS price computed at $N = 256$, and choose N as the smallest value at which the error reaches the floor set by the trapezoidal payoff quadrature.

(a) Both underlyings log–normal

In this case, both state variables are the log-prices $x_i = \ln S_i(T)$. Under $\mathbb{Q}$,

$$S_i(T) = S_i(0)\exp\big((r - \tfrac{1}{2}\sigma_i^2)T + \sigma_i\sqrt{T}\,Z_i\big),$$

so $x_i \sim N(\mu_i, \sigma_i^2 T)$ with $\mu_i = \ln S_i(0) + (r - \frac{1}{2}\sigma_i^2)T$, and the normal characteristic function is

$$\varphi_i(\omega) = \exp\big(i\omega\mu_i - \tfrac{1}{2}\sigma_i^2 T\,\omega^2\big), \qquad i = 1, 2.$$

The payoff map is $g_i(x) = e^x$. The Gaussian characteristic function decays like $e^{-\frac{1}{2}\sigma_i^2 T\,\omega^2}$, so the density coefficients decay extremely fast and exponential convergence is expected.

For the truncation, notice that the normal has cumulants $\kappa_1 = \mu_i$, $\kappa_2 = \sigma_i^2 T$, $\kappa_3 = \kappa_4 = 0$, giving $x_1 \in [2.205, 7.005]$ and $x_2 \in [0.980, 8.180]$.

After running our code, we get that the converged COS price is 8.16827. Figure 10 shows the error against the converged price falling by a growing factor at each step (18.6×, 32.2×, 77.3×, 208.3× per step), which is faster than any algebraic rate, until it saturates near 10^{-8}, the trapezoidal quadrature floor of the payoff coefficients. $N = 48$ already reaches this floor (Figure 11).

```
  truncation domains: x1 in [2.205, 7.005], x2 in [0.980, 8.180]
  converged COS price (N = 256): 8.168268
   N     COS price      |error|      ratio
------------------------------------------
   8    -8.544936    1.67e+01
  16     7.271769    8.96e-01      18.6x
  24     8.140446    2.78e-02      32.2x
  32     8.167909    3.60e-04      77.3x
  40     8.168267    1.73e-06     208.3x
  48     8.168268    1.65e-08     104.6x
  64     8.168268    1.95e-08       0.8x
```

Figure 10: COS price and error against the converged COS price 8.16827.

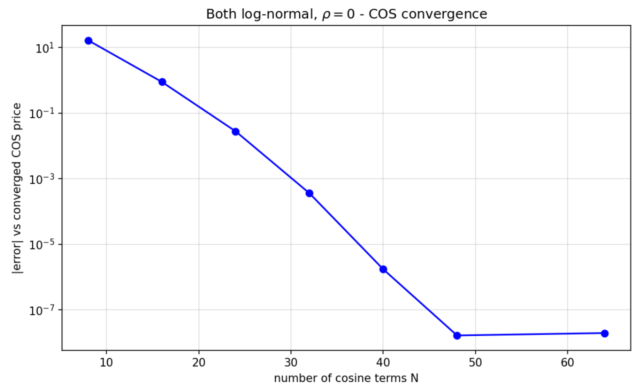

Figure 11: Absolute error of the COS price versus the number of cosine terms N. The error decays exponentially until the quadrature floor near 10^{-8}.

(b) One log–normal, one CIR

We observe that the first state variable is again the log-price $x_1 = \ln S_1(T) \sim N(\mu_1, \sigma_1^2 T)$ with the normal characteristic function above ($\sigma_1 = 0.25$). For the second we keep S_2 $y_2 = S_2(T)$, and we know its law. Under $\mathbb{Q}$ the second asset is the square-root process $dS_2 = rS_2\,dt + \sigma_2\sqrt{S_2}\,dW_2$; the physical mean-reversion parameters $\kappa = 0.015$ and $\theta = 100$ drop out under the risk-neutral measure (Section 1), leaving only $\sigma_2 = 1.5$. By the technical hint given in the beginning of the assignment,

$$S_2(T) = c\,Z, \quad c = e^{rT}\,\frac{\sigma_2^2\tau(T)}{4}, \quad \tau(T) = \frac{1-e^{-rT}}{r}, \quad Z \sim \chi_0'^2(\lambda), \quad \lambda = \frac{4S_2(0)}{\sigma_2^2\tau(T)}.$$

The characteristic function of a non-central chi-square $\chi_d'^2(\lambda)$ with d degrees of freedom is $(1-2iu)^{-d/2}\exp\bigl(\lambda iu/(1-2iu)\bigr)$. For $d=0$, $(1-2iu)^{-d/2}$ equals 1, giving $\varphi_Z(u) = \exp\bigl(\lambda iu/(1-2iu)\bigr)$, and therefore

$$\varphi_2(\omega) = \mathbb{E}\bigl[e^{i\omega c Z}\bigr] = \varphi_Z(\omega c) = \exp\Bigl(\lambda\,\frac{i(\omega c)}{1-2i(\omega c)}\Bigr).$$

The payoff map is the identity $g_2(y_2) = y_2$ (the state already is the price), while $g_1(x_1) = e^{x_1}$.

Note that the cumulants of Z follow from its cumulant generating function:

$$\log\varphi_Z(u) = \lambda\sum_{m\geq 1} 2^{m-1}(iu)^m \quad\Rightarrow\quad \kappa_m(Z) = 2^{m-1}m!\,\lambda,$$

therefore for $S_2 = cZ$, we get

$$\kappa_m(S_2) = c^m\,2^{m-1}m!\,\lambda, \qquad \kappa_1 = c\lambda, \quad \kappa_2 = 4c^2\lambda, \quad \kappa_4 = 192\,c^4\lambda.$$

The rule gives $S_2 \in [-103.0, 307.0]$, clipped to $[0, 307.0]$ since $S_2 \geq 0$ is a price and cannot be negative, and $x_1 \in [1.594, 7.594]$.

After running our code, the converged COS price is 6.78259. Figure 12 shows the same quasi-exponential decay (error falling by factors 18.7, 31.3, 76.5 and 197.2 at successive steps) down to the quadrature floor near 10^{-8}, reached by $N = 48$ (Figure 13).

```
truncation domains: x1 in [1.594, 7.594], S2 in [0.000, 307.004]
converged COS price (N = 256): 6.782585
 N    COS price      |error|      ratio
----------------------------------------
  8     0.631716    6.15e+00
 16     6.453013    3.30e-01      18.7x
 24     6.772042    1.05e-02      31.3x
 32     6.782447    1.38e-04      76.5x
 40     6.782584    6.99e-07     197.2x
 48     6.782585    2.96e-08      23.6x
 64     6.782585    2.84e-08       1.0x
```

Figure 12: COS price and error against the converged COS price 6.78259.

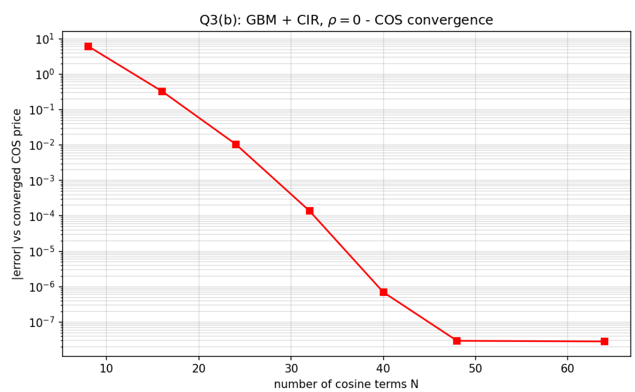

Figure 13: Absolute error of the COS price versus the number of cosine terms N. The error decays exponentially until the quadrature floor near 10^{-8}.